



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP3723 SYSTEM DEVELOPMENT TECHNOLOGY
SEMESTER 1 2024/2025
SECTION 1

Individual Test: Course Registration System

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System Introduction

The course registration system is the individual project assigned for this subject, System Development & Technology (SECP3723). In this individual project, we have applied a few programming languages which are specialised in web application development such as PHP, CSS, JavaScript and HTML. By applying this knowledge learnt along the semester, a web-based course registration system was developed which provides students a platform to register for courses for upcoming semesters.

The course registration system has 3 types of users, which are admin, student and lecturer. The student is the user that can register for courses, view current and previous semester courses, search courses, modify their registration, cancel their registration, and edit their profile. The lecturer on the other hand can view their assigned courses for each semester, view student lists for each course they are assigned to, view course details and view student details. Finally, the admin is responsible to add new course details with the number of maximum students per course, modify existing course details, delete existing courses and amend or cancel registration.

System Design

Entity Relationship Diagram(ERD)

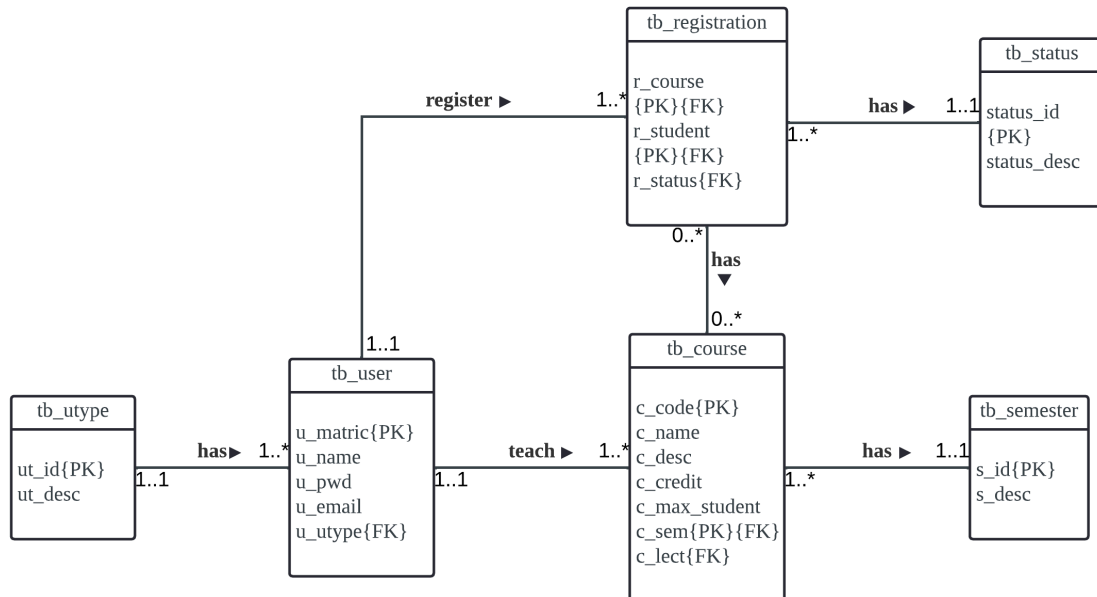


Figure 1.1 ERD of Course Registration System

Use Case Diagram

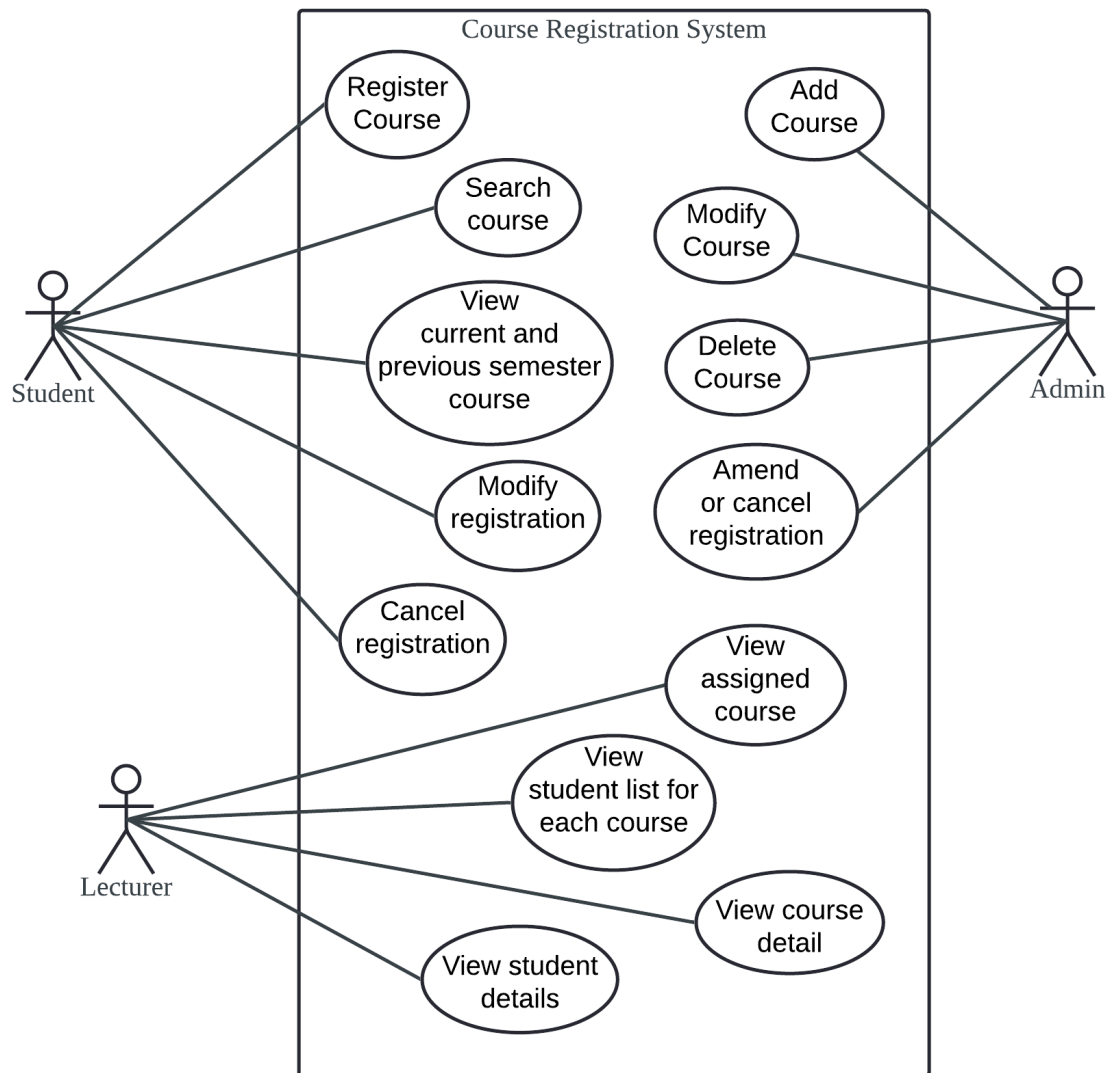


Figure 1.2 Use Case Diagram of Course Registration System

Development Software, Language, Technology and Tools

Software

The softwares that was used in this individual project to complete the car booking system are Visual Studio Code, XAMPP and Google browser. Visual Studio Code is a source code editor which supports a variety of programming and markup languages through the extensions that were provided for installation. Other than that, XAMPP is an open-source cross-platform web server solution stack package which is able to support me to host the course registration locally and offline by using Apache HTTP server. Lastly, the Google Chrome browser was used to test the functionalities of the system.

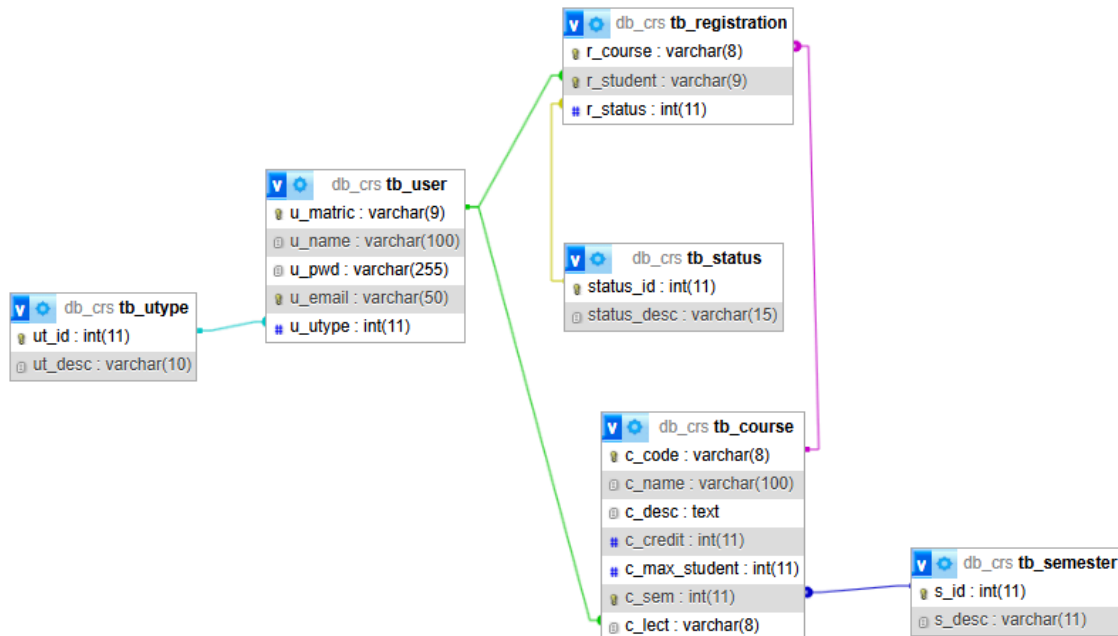
Languages

There are many types of programming languages that was was used to the develop the course registration system which were Hypertext Markup Language (HTML), Cascading Style Sheets (CSS), JavaScript, Hypertext Preprocessor (PHP) and Structured Query Language (SQL). HTML is a programming language that is used to build and structure web pages with the help of CSS to define the styles of the web pages. Other than that, JavaScript is a scripting language for web pages which use to trigger functions while certain events occur. PHP is a general proposed scripting language that connects the HTML to the database which is phpMyAdmin. Lastly, SQL language is also used in the phpMyAdmin to administrate the MySQL database of the system.

Tools

The tool used was Lucidchart which is an online tool that is used to construct the ERD and use case diagram of the system.

Database Design



Server: 127.0.0.1 » Database: db_crs » Table: tb_course

Browse Structure SQL Search Insert Export Import Privileges Operatic

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	c_code 🔑	varchar(8)	utf8mb4_general_ci		No	None			🔧 Change 🛑 Drop 🔍 More
☐ 2	c_name	varchar(100)	utf8mb4_general_ci		No	None			🔧 Change 🛑 Drop 🔍 More
☐ 3	c_desc	text	utf8mb4_general_ci		No	None			🔧 Change 🛑 Drop 🔍 More
☐ 4	c_credit	int(11)			No	None			🔧 Change 🛑 Drop 🔍 More
☐ 5	c_max_student	int(11)			No	None			🔧 Change 🛑 Drop 🔍 More
☐ 6	c_sem 🔑🔑	int(11)			No	None			🔧 Change 🛑 Drop 🔍 More
☐ 7	c_lect 🔑	varchar(8)	utf8mb4_general_ci		No	None			🔧 Change 🛑 Drop 🔍 More

Server: 127.0.0.1 » Database: db_crs » Table: tb_registration

Browse Structure SQL Search Insert Export Import Privileges Oper

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	r_course 🔑🔑	varchar(8)	utf8mb4_general_ci		No	None			🔧 Change 🛑 Drop 🔍 More
☐ 2	r_student 🔑🔑	varchar(9)	utf8mb4_general_ci		No	None			🔧 Change 🛑 Drop 🔍 More
☐ 3	r_status 🔑	int(11)			No	None			🔧 Change 🛑 Drop 🔍 More

Server: 127.0.0.1 » Database: db_crs » Table: tb_semester

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[Privileges](#)
[Operations](#)

[Table structure](#)
[Relation view](#)

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	s_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	s_desc	varchar(11)	utf8mb4_general_ci		No	None			Change Drop More

Server: 127.0.0.1 » Database: db_crs » Table: tb_status

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[Privileges](#)
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#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	status_id	int(11)			No	None			Change Drop More
☐ 2	status_desc	varchar(15)	utf8mb4_general_ci		No	None			Change Drop More

Server: 127.0.0.1 » Database: db_crs » Table: tb_user

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[Relation view](#)

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	u_matric	varchar(9)	utf8mb4_general_ci		No	None			Change Drop More
☐ 2	u_name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	u_pwd	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐ 4	u_email	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐ 5	u_utype	int(11)			No	None			Change Drop More

Server: 127.0.0.1 » Database: db_crs » Table: tb_utype

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#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	ut_id	int(11)			No	None			Change Drop More
☐ 2	ut_desc	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More

Development Steps

Phase 1: Planning and Requirements Gathering.

In this phase, research was done to identify the requirements of the building. Questions like, “What type of data is involved in the system?” and “What sort of functionalities are needed to be in the system?” These sorts of questions help determine the system's purpose, objective, target audience, and range. The phase focuses on how things are done, identifying the proper flow of information throughout the system.

Phase 2: Designing the system

The system's ERD and use case diagram were first created. The ERD provides a visualization of the data needed in the system that also determines the logic and business rules of the system. The use case diagrams explain the context and requirements of the system showing ways in which a user interacts with the system. It also determines the scope of the system showing functionalities that are not covered by the system.

Phase 3: Developing the system

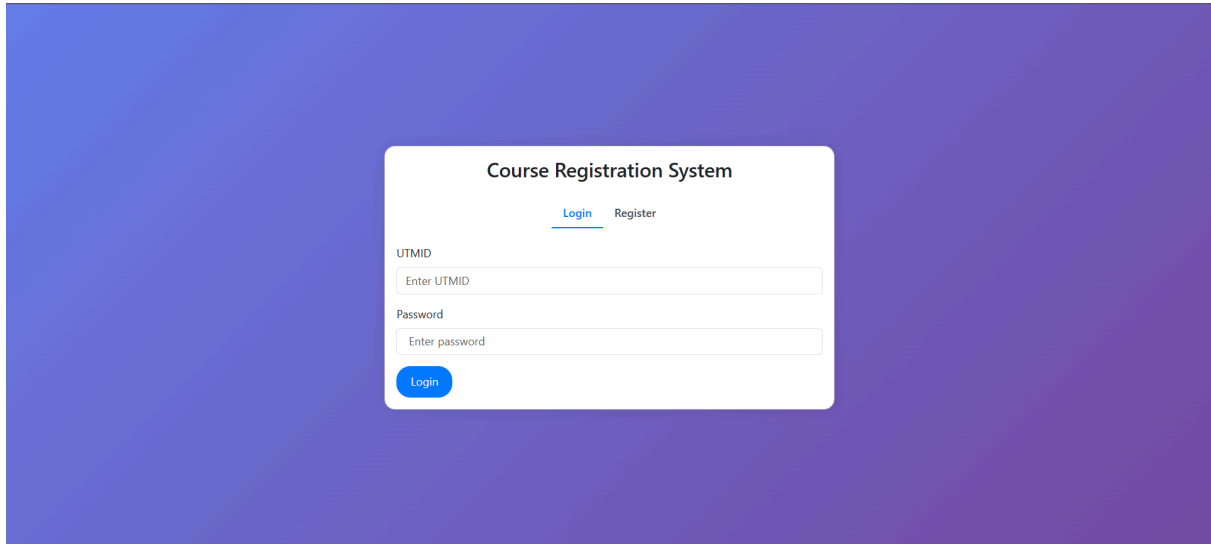
The system was developed with various programming languages that are used to create a functioning client side interface for presentation that is interactive while also having a server side that controls all the operations within the system such as CRUD operations, password encryption, SQL injection protection, password validation and resolving login error.

Phase 4: Testing the system

The testing phase was done each time a module or a use case was done. This phase is important to identify logical errors and bugs to ensure they can be fixed so that the system can work as intended. The testing was done on the web server each time the code was finished implementing. Then, an integration test was done by simulating the whole flow of the system to identify incompatibility issues for all the code that was developed.

System Interface

Login Interface for all user



The login interface is a white card centered on a purple gradient background. It features a title 'Course Registration System' and two tabs: 'Login' (active) and 'Register'. Below the tabs are two input fields: 'UTMID' with placeholder text 'Enter UTMID' and 'Password' with placeholder text 'Enter password'. A blue 'Login' button is positioned at the bottom left of the card.

Course Registration System

[Login](#) [Register](#)

UTMID

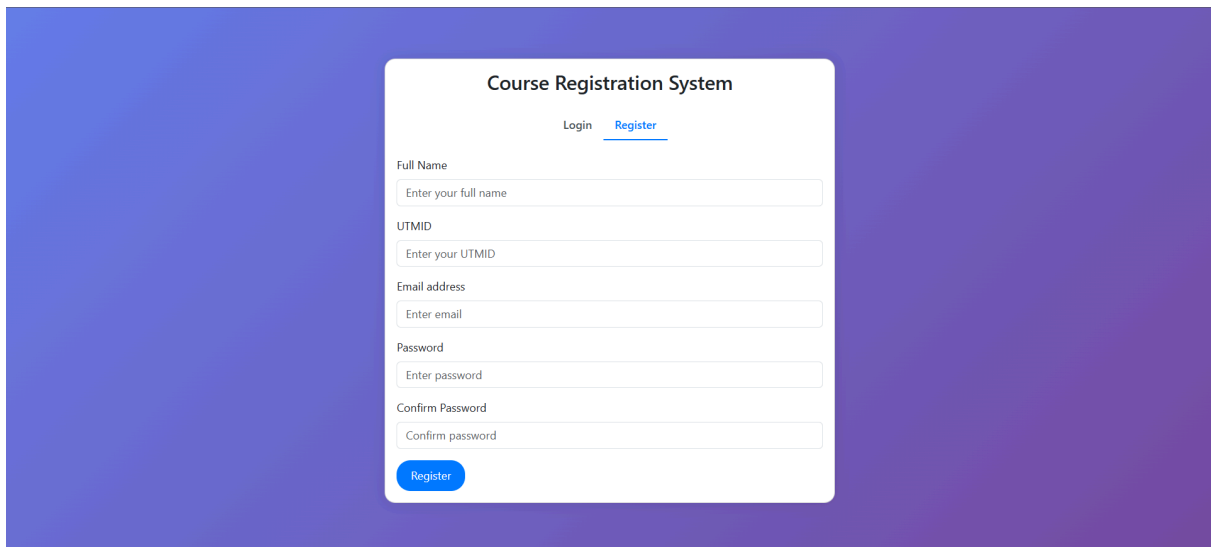
Enter UTMID

Password

Enter password

Login

Registration Interface for Students



The registration interface is a white card centered on a purple gradient background. It features a title 'Course Registration System' and two tabs: 'Login' and 'Register' (active). Below the tabs are five input fields: 'Full Name' with placeholder text 'Enter your full name', 'UTMID' with placeholder text 'Enter your UTMID', 'Email address' with placeholder text 'Enter email', 'Password' with placeholder text 'Enter password', and 'Confirm Password' with placeholder text 'Confirm password'. A blue 'Register' button is positioned at the bottom left of the card.

Course Registration System

[Login](#) [Register](#)

Full Name

Enter your full name

UTMID

Enter your UTMID

Email address

Enter email

Password

Enter password

Confirm Password

Confirm password

Register

Student

Registered Course Page

Course Registration SystemRegistered CourseRegister CourseEdit ProfileLogout

Registered Courses

Semester

2023/2024-1

Submit

No.	Course Code	Course Name	Credit	Action
1	SECJ1013	Programming Technique 1	3	Remove
2	SECR1013	Digital Logic	3	Remove

Course Registration Page

Course Registration SystemRegistered CourseRegister CourseEdit ProfileLogout

Course Registration

Semester

2023/2024-1

Submit

Search course...

Search

No.	Course Code	Course Name	Credit	Capacity	Semester	Lecturer	
1	SECJ1013	Discrete Structure	3	1/2	2023/2024-1	Dr Shakir	Register
2	SECJ1013	Programming Technique 1	3	1/30	2023/2024-1	Dr Shakir	Register
3	SECP1513	Technology and Information System	3	0/30	2023/2024-1	Dr Alimi	Register
4	SECR1013	Digital Logic	3	1/30	2023/2024-1	Dr Alimi	Register

Edit Profile Page

Course Registration System Registered Course Register Course Edit Profile Logout

Student Profile

UTMID

100

Name

Iman Abadi

Email

imanabadi@graduate.utm.my

Current Password

Enter current password

New Password

Enter new password

Confirm Password

Confirm password

Update

Lecturer

Assigned Course Page

Assigned Courses

No.	Course Code	Course Name	Credit	Students		
1	SECI1013	Discrete Structure	3	1/2	Details	Students
2	SECI1013	Programming Technique 1	3	1/30	Details	Students

Course Detail Page

Course Detail

SECI1013 - Discrete Structure

Course Description:

This course introduces students to the principles and applications of discrete structure in the field of computer science. The topics that are covered in this course are set theory, proof techniques, relations, functions, recurrence relations, counting methods, graph theory, trees and finite automata. At the end of the course, the students should be able to use set theory, relations and functions to solve computer science problems, analyze and solve problems using recurrence relations and counting methods, apply graph theory and trees in real world problems and use deterministic finite automata finite state machines to model electronic devices and problems.

Back

Student List of Assigned Course Page

Course Registration SystemAssigned CourseStudent ListLogout

Student List

No.	Matric Number	Name
1	101	Ali Abdul

Back

Student List Page

Course Registration SystemAssigned CourseStudent ListLogout

Student List

No.	Matric Number	Name	
1	100	Iman Abadi	View
2	101	Ali Abdul	View
3	102	Amirul	View

Student Detail Page

[Course Registration System](#) [Assigned Course](#) [Student List](#) [Logout](#)

Student Details

Name: Iman Abadi

Matric Number: 100

Email: imanabadi@graduate.utm.my

[Back](#)

Admin

Admin Manage Course Page

Course Registration System

Course

Manage Registration

Logout

Courses

Add Course

No.	Course Code	Course Name	Credit	Capacity	Semester Offered	Lecturer	Actions
1	SEC1013	Discrete Structure	3	2	2023/2024-1	Dr Shakir	EditDelete
2	SEC1013	Programming Technique 1	3	30	2023/2024-1	Dr Shakir	EditDelete
3	SEC1023	Programming Technique II	3	30	2023/2024-2	Dr Alimi	EditDelete
4	SECP1513	Technology and Information System	3	30	2023/2024-1	Dr Alimi	EditDelete
5	SECR1013	Digital Logic	3	30	2023/2024-1	Dr Alimi	EditDelete
6	SECV2113	Human Computer Interaction	3	30	2023/2024-2	Dr Alimi	EditDelete

Admin Add Course Page

Course Registration System

Course

Manage Registration

Logout

Course Details

Code

Enter course code

Name

Enter course name

Description

Credit Hour

Enter credit hour

Capacity

Enter capacity

Semester

Admin Edit Course Page

Course Registration System Course Manage Registration Logout

Course Details

Code

SECI1013

Name

Discrete Structure

Description

This course introduces students to the principles and applications of discrete structure in the field of computer science. The topics that are covered in this course are set theory, proof techniques, relations, functions, recurrence relations, counting methods, graph theory, trees and finite automata. At the end of the course, the students should be able to use set theory, relations and functions to solve computer science problems, analyze and solve problems

Credit Hour

3

Capacity

2

Semester

Admin Amend Registration Page

Course Registration System Course Manage Registration Logout

Student List

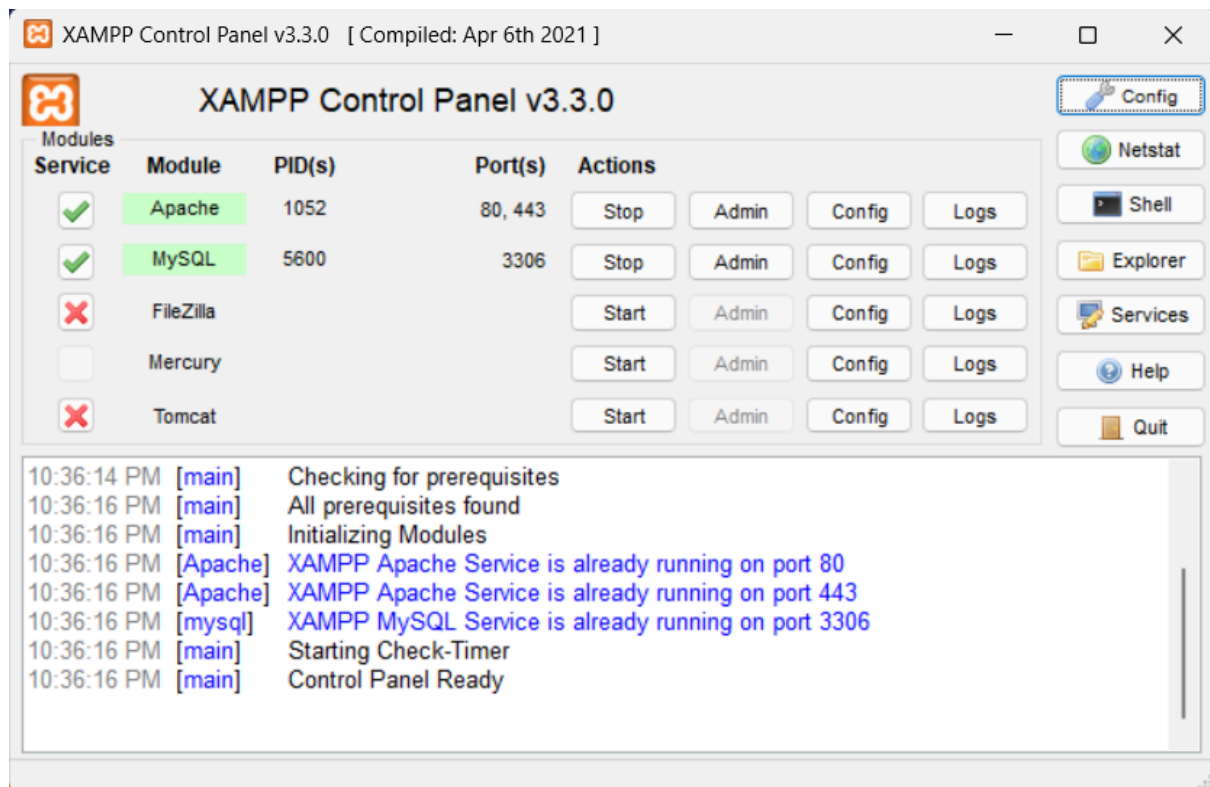
No.	Matric Number	Name	Course Code	Course Name	Status	
1	102	Amirul	SECI1013	Discrete Structure	Sent	ApproveReject

Localhost Setup

1. Download and install XAMPP Control Panel from

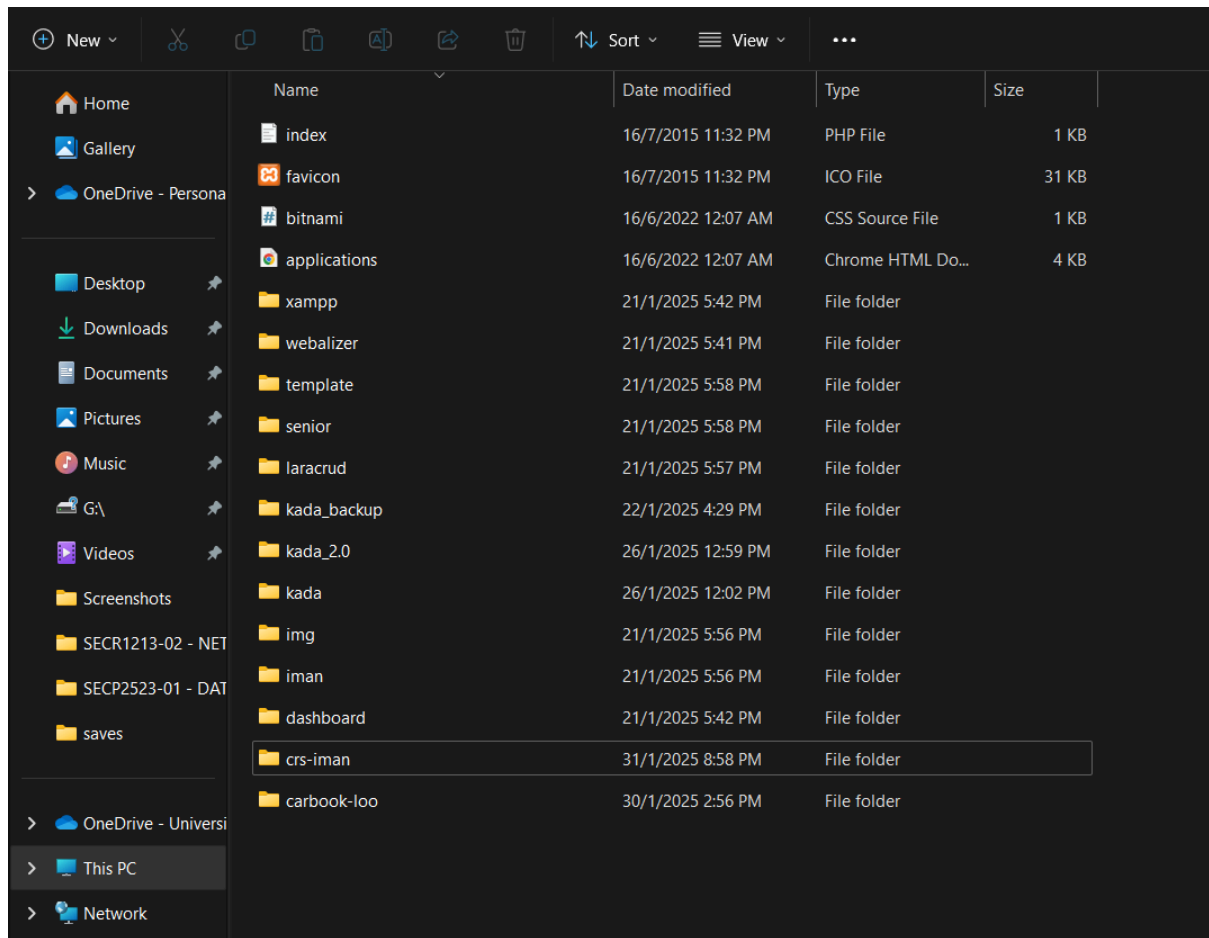
<https://www.apachefriends.org/index.html>

2. Start Apache and MySQL.

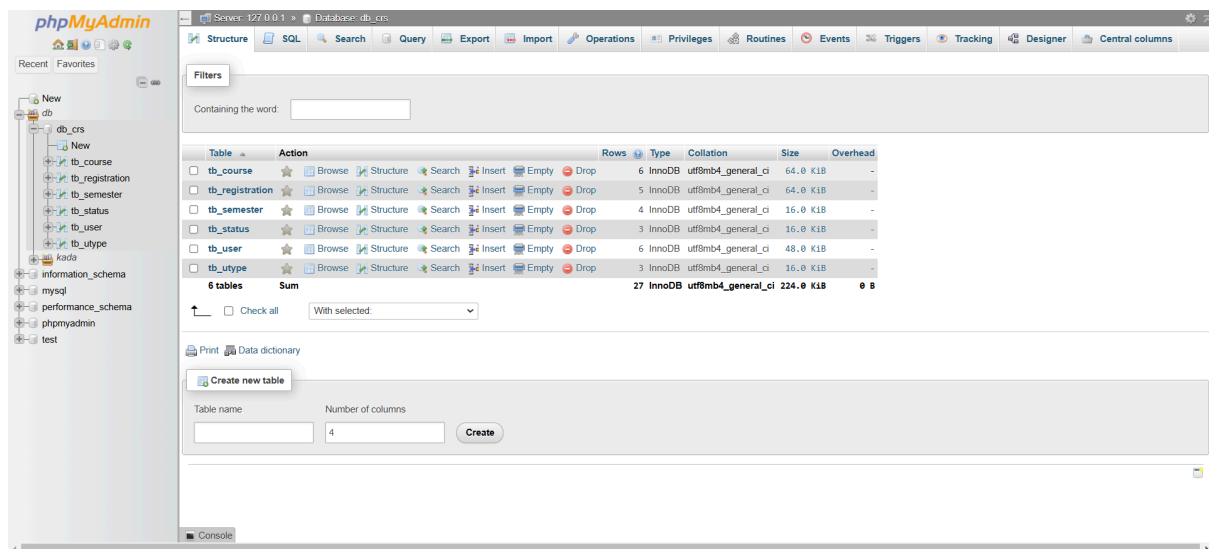


3. View phpMyAdmin by clicking 'Admin' button on 'Actions' section of MySQL module.

4. Extract the 'crs-iman' file into xampp > htdocs file path.



5. Create a new database with name 'db_crs'



User Credentials

Student Accounts

UTMID: 100

Password: 1234

UTMID: 101

Password: 1234

UTMID: 102

Password: 1234

Lecturer Accounts

UTMID: 200

Password: 1234

UTMID: 201

Password: 1234

Admin Accounts

UTMID: 300

Password: 1234